MASTER OF SCIENCE THESIS DEFENSE

# On the Robustness of Decentralized Federated Learning against Byzantine Attacks and Channel Noise

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# The thesis in one slide

## Research question

How can a receiver learn reliably from peers under attacks, channel noise and label skew?

## Methodology

FedSharp: guarded exchange + noise-aware training.
FedSharp+: class-aware admission + update checks.

## Result

FedSharp+: 28.98% on CIFAR-10 non-IID joint cases.
No gain over FedSharp on MNIST.

## Core references

[1] McMahan et al., AISTATS 2017
Federated averaging
[2] Elkordy et al., IEEE JSAC 2022
BASIL

# Collaborative learning has three practical constraints

### Local data

Clients keep their fitting examples.

### No central server

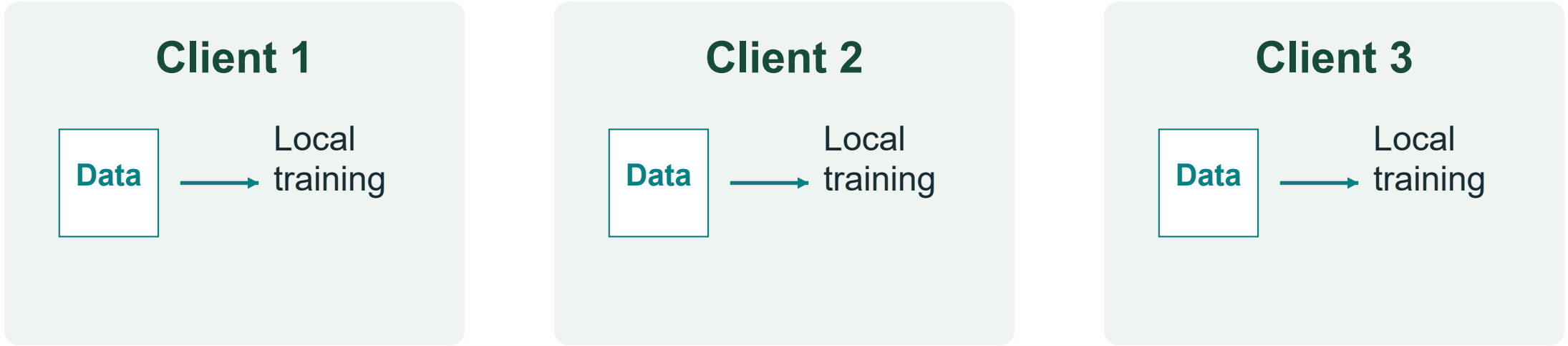
Peers exchange model parameters.

### Unreliable exchange

Malicious updates and noisy links.

**Removing the server does not remove the need to decide which model to trust.**

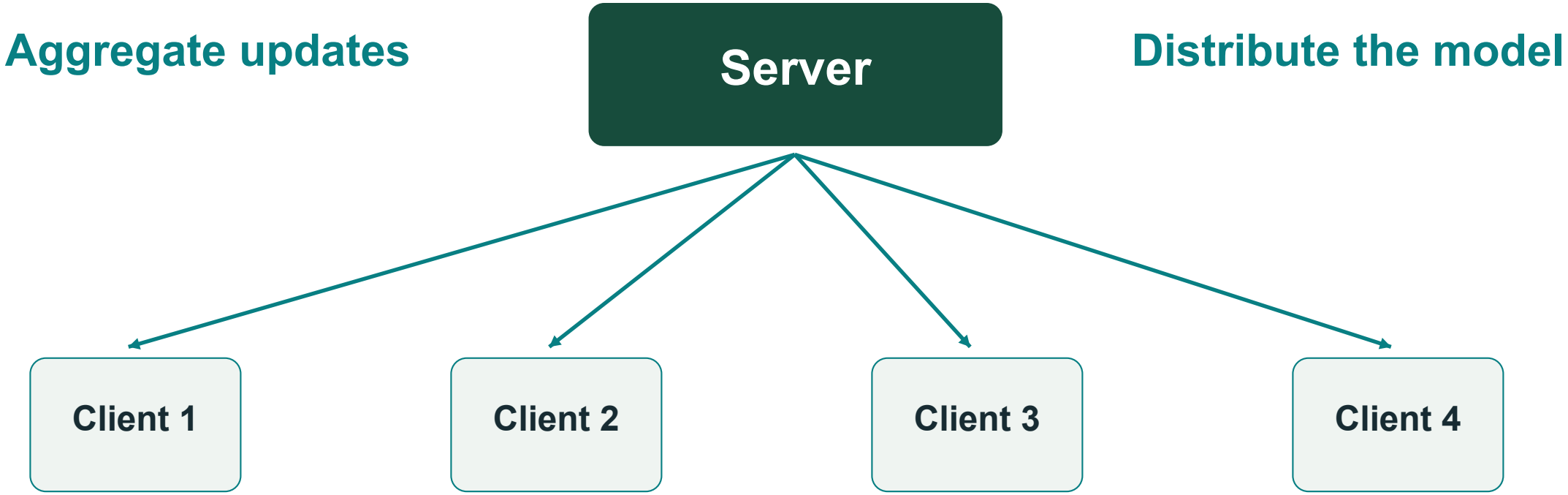
# Federated learning keeps training data at the clients




## Exchange model parameters

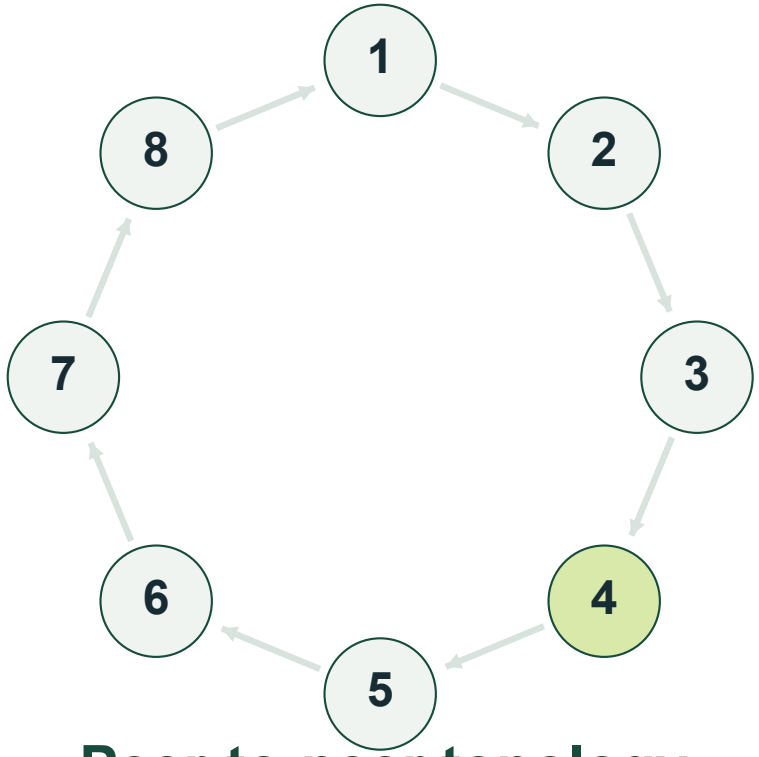
**The data remain local; the models connect the clients.**

# Centralized federated learning uses a coordinating server




**Clients train locally, but coordination still depends on one server.**

# Decentralized FL moves model exchange to the peers




**Peer-to-peer topology**

## What changes?

No central aggregator.
A receiver makes its own decision.
Local data may differ across clients.

**This thesis studies a sequential directed ring within the broader DFL setting.**

# BASIL provides the starting point for the ring study

### Byzantine resilience

BASIL evaluates received models on the receiver's local data. [2]

### Statistical heterogeneity

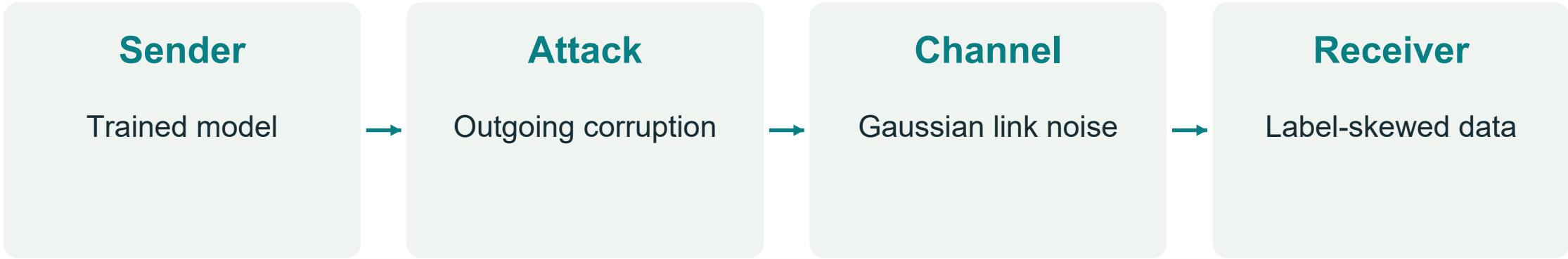
Non-IID data can make local model preferences disagree. [3]

### Communication noise

Link perturbations can accumulate during repeated exchange. [4]

**Study attacks and channel noise together, then address receiver-level class recognition.**

# One model can carry several kinds of error




**The receiver sees the combined effect, not a label identifying the source of error.**

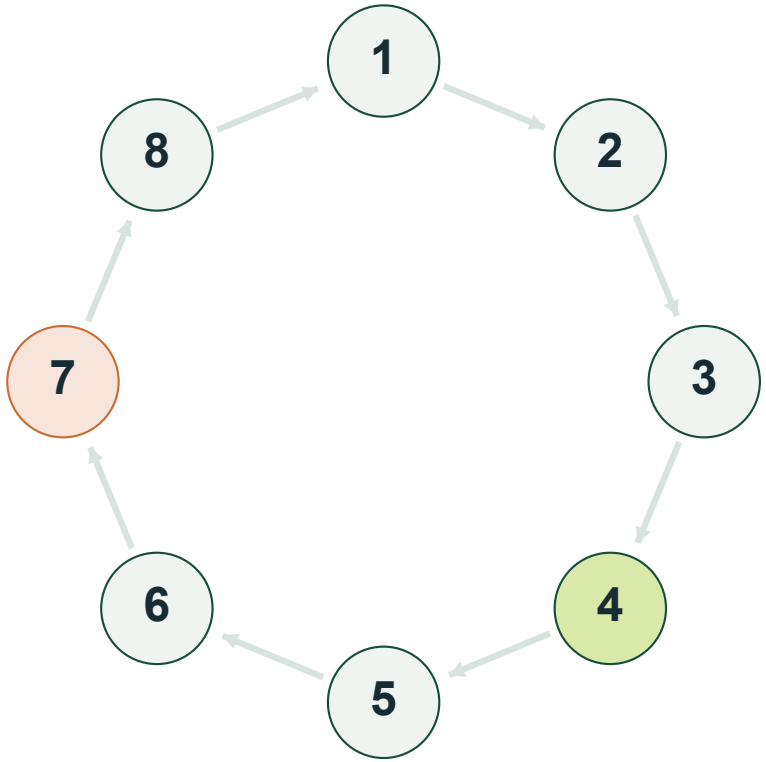
# Three research questions guide the evaluation

1 **RQ1** Can guarded exchange and noise-aware training improve joint attack and noise performance?

2 **RQ2** Can receiver-local class evidence improve learning on non-IID data without sharing examples?

3 **RQ3** Where do these methods help, and what performance costs remain?

# Each receiver uses a memory of predecessor models




Schematic: 8 of 100 clients

**Memory**

Received versions

→

**Decision**

Select or admit

→

**Update**

Five SGD steps

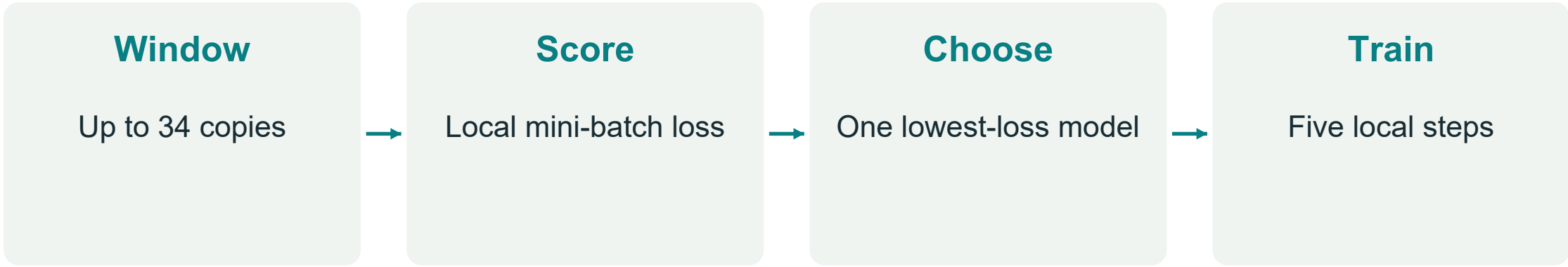
**BASIL: up to 34 copies**
**FedSharp: up to 66 copies**

# The evaluation follows a fixed, auditable pipeline



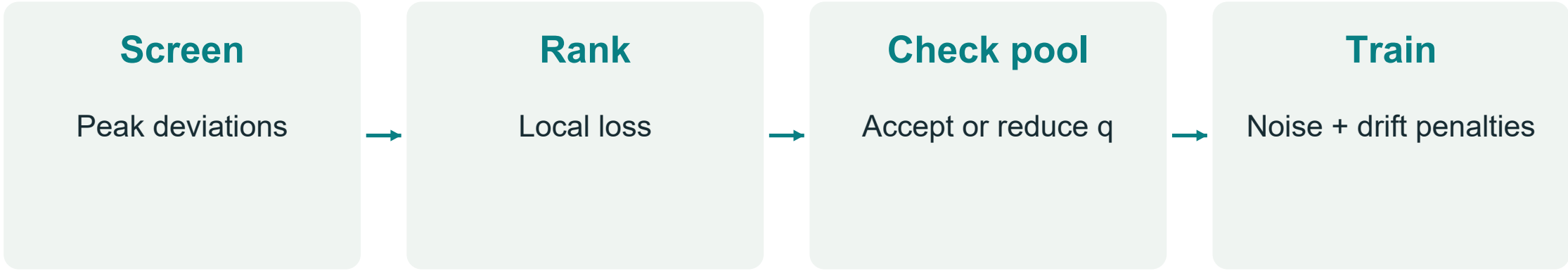

**Match the data and training budget before comparing the methods.**

# BASIL forwards one locally preferred candidate




**A low local loss may not reflect recognition across all classes.**

# FedSharp adds safeguards before and during training




## What enters local training?

An accepted parameter vector, after the receiver checks the realized average.

# FedSharp averages model parameters, not the lowest losses

$$\bar{\boldsymbol{\theta}}_i = \frac{1}{|\mathcal{Q}_i|} \sum_{s \in \mathcal{Q}_i} \tilde{\boldsymbol{\theta}}_{s,i}$$

### Rank parameters

Use local loss values

→

### Test their average

Check the realized loss

→

### Reduce if needed

Actual count q can be 1

**The maximum budget is 33; the accepted count is a measured outcome.**

# The training objective addresses noise and local drift

$$\ell_{i,k}^{\text{robust}}(\boldsymbol{\theta};\zeta) = \ell_i(\boldsymbol{\theta};\zeta) + \lambda_k \left|\nabla_{\boldsymbol{\theta}} \ell_i(\boldsymbol{\theta};\zeta)\right|_2^2$$

$$\mathcal{L}_{i,k}(\boldsymbol{\theta};\zeta) = \ell_{i,k}^{\text{robust}}(\boldsymbol{\theta};\zeta) + \frac{\mu}{2}\left|\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}_i\right|_2^2$$

## Gradient penalty

Reduce local loss sensitivity.

## Proximal anchor

Limit movement from initialization.

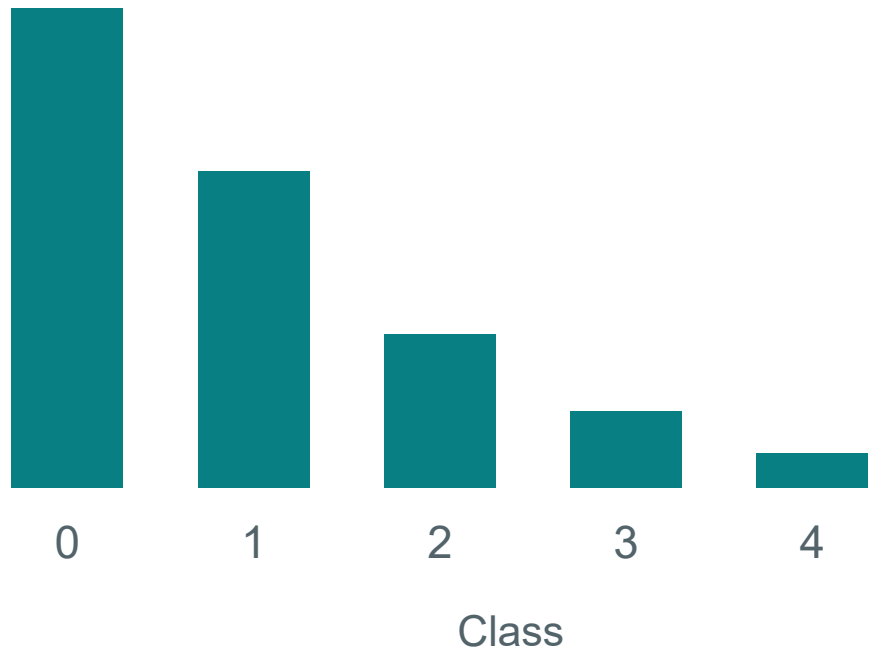
$$\lambda_k = \frac{\sigma^2}{2q\eta_k h}$$

q is the realized count in code.

Thesis robust and proximal objectives, unchanged. Proximal training: Li et al. [5].

# Under label skew, one loss value may hide class failures

## Illustrative local label distribution




### Receiver-local question

Does this candidate preserve the classes this receiver can already recognize?

Schematic only. Class counts are illustrative, not experimental measurements.

# FedSharp+ admits models using local class evidence




## Little class support

Use the checked pool

## Uninformative owned model

Prefer the nearest candidate

## Informative owned model

Compare class-aware utility

# FedSharp+ checks the whole update before sending it

**Train**

Local prior + five SGD steps

→

**Repair**

Classifier, when eligible

→

**Commit**

Whole-model scale check

$$\theta_i(s) = a_i + s(\theta_i^+ - a_i), \qquad s \in \{1, \tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, 0\}$$

$$Q_i(\theta_i(s)) \geq Q_i(a_i) - \delta$$

**Accept the full update, a smaller update, or the admitted model.**

Training-time class-prior adjustment follows Menon et al. [6]. Utility checks are local, not population guarantees.

EXPERIMENTAL DESIGN

# The study covers two datasets and three independent seeds

**100**
clients

**500**
rounds

**5**
SGD steps / activation

**3**
seeds

## Data and partitions

MNIST + CIFAR-10
IID + Dirichlet non-IID (alpha = 0.5)

## Evaluation

1,000-image development holdout
Endpoint: rounds 491 to 500, mean ± sample SD

# Clean controls establish the transfer baseline

## Clean environment

No attack or channel noise.
Accept the immediate predecessor.
No selection or regularization.

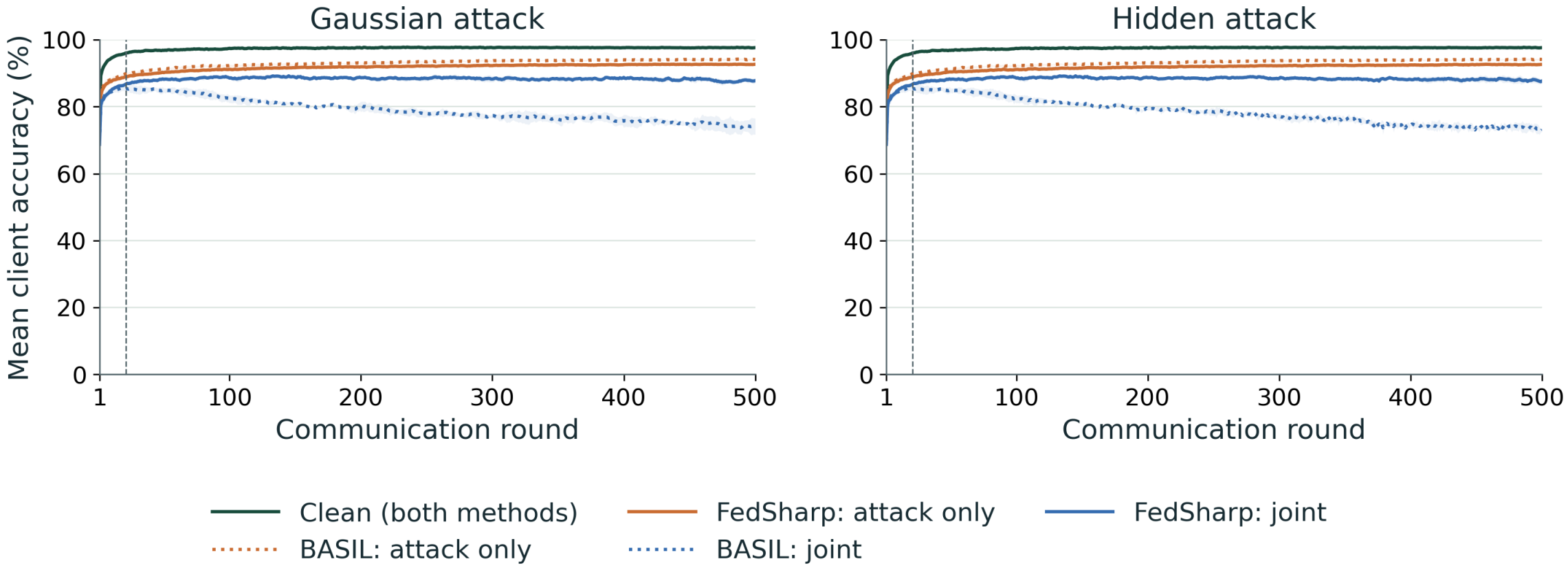
## Impaired environment

Attack only, or attack + noise.
Selection and defenses remain enabled.
FedSharp+ is tested on non-IID joint cases.

Joint tests: 33 attackers | Noise SD = 0.01 | Attack onset: round 21

**BASIL and FedSharp clean accuracy arrays match exactly in all 12 pairs.**

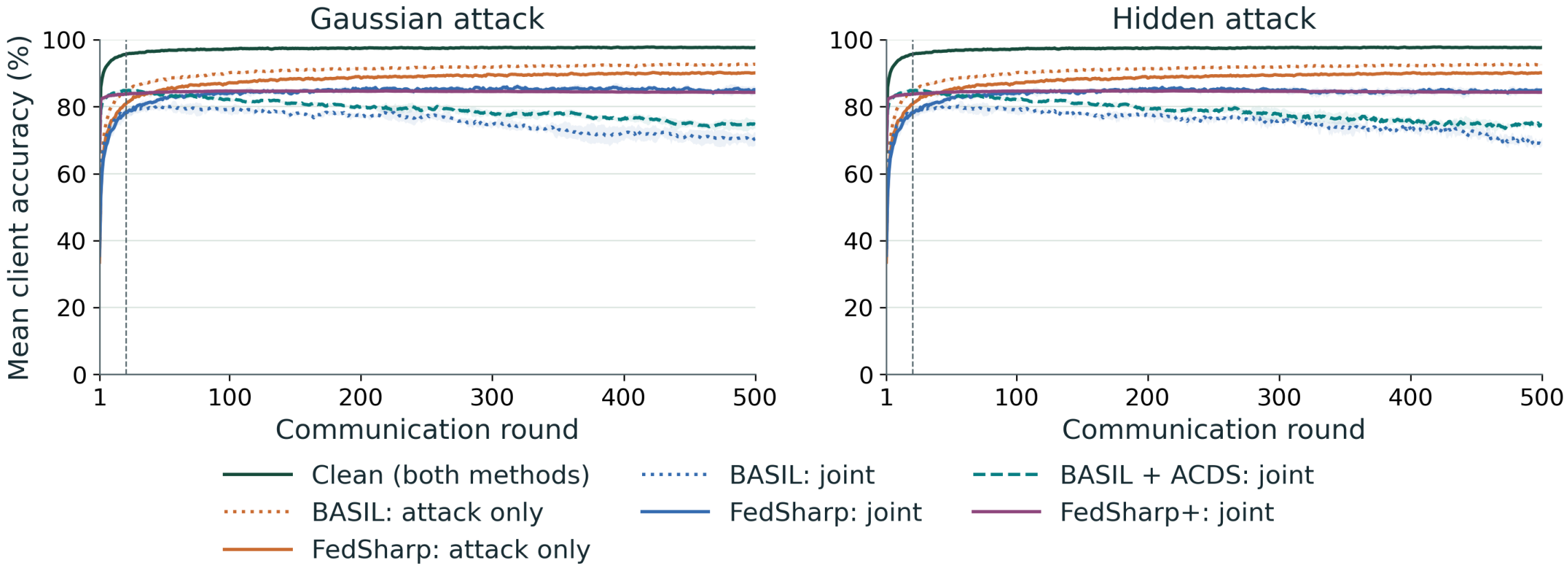
# MNIST IID: accuracy through 500 rounds




**FedSharp improves the joint-impaired result.**

Joint = attack + noise. Three seeds; bands = sample SD. Clean curves coincide. Development-holdout accuracy.

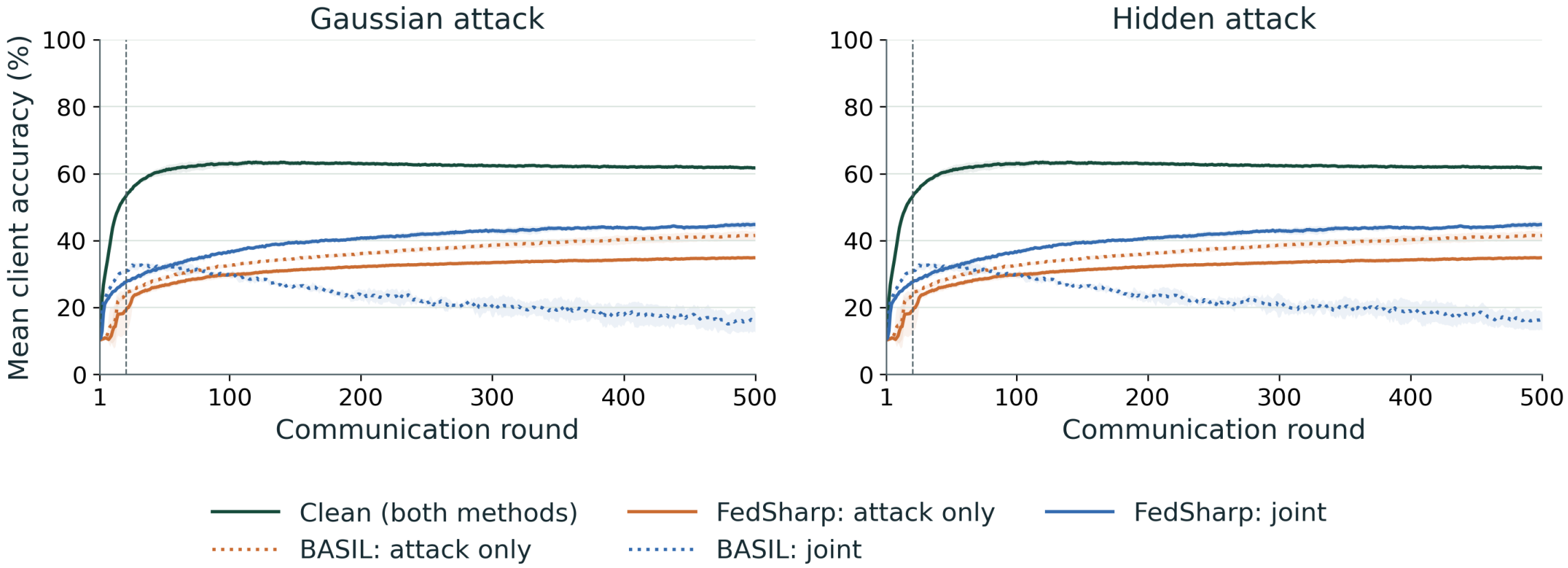
# MNIST non-IID: accuracy through 500 rounds




**FedSharp+ does not improve the MNIST mean over FedSharp.**

Joint = attack + noise. Three seeds; bands = sample SD. Clean curves coincide. Development-holdout accuracy.

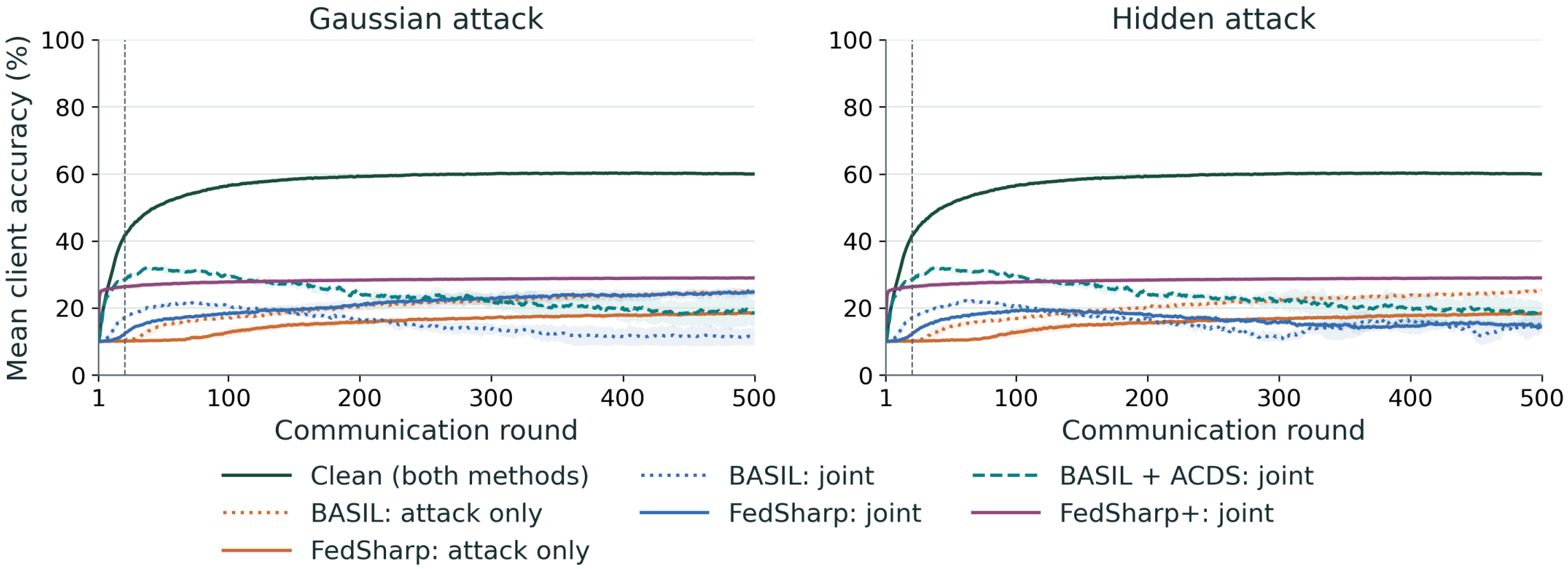
# CIFAR-10 IID: accuracy through 500 rounds




**FedSharp helps under joint impairments; clean remains highest.**

Joint = attack + noise. Three seeds; bands = sample SD. Clean curves coincide. Development-holdout accuracy.

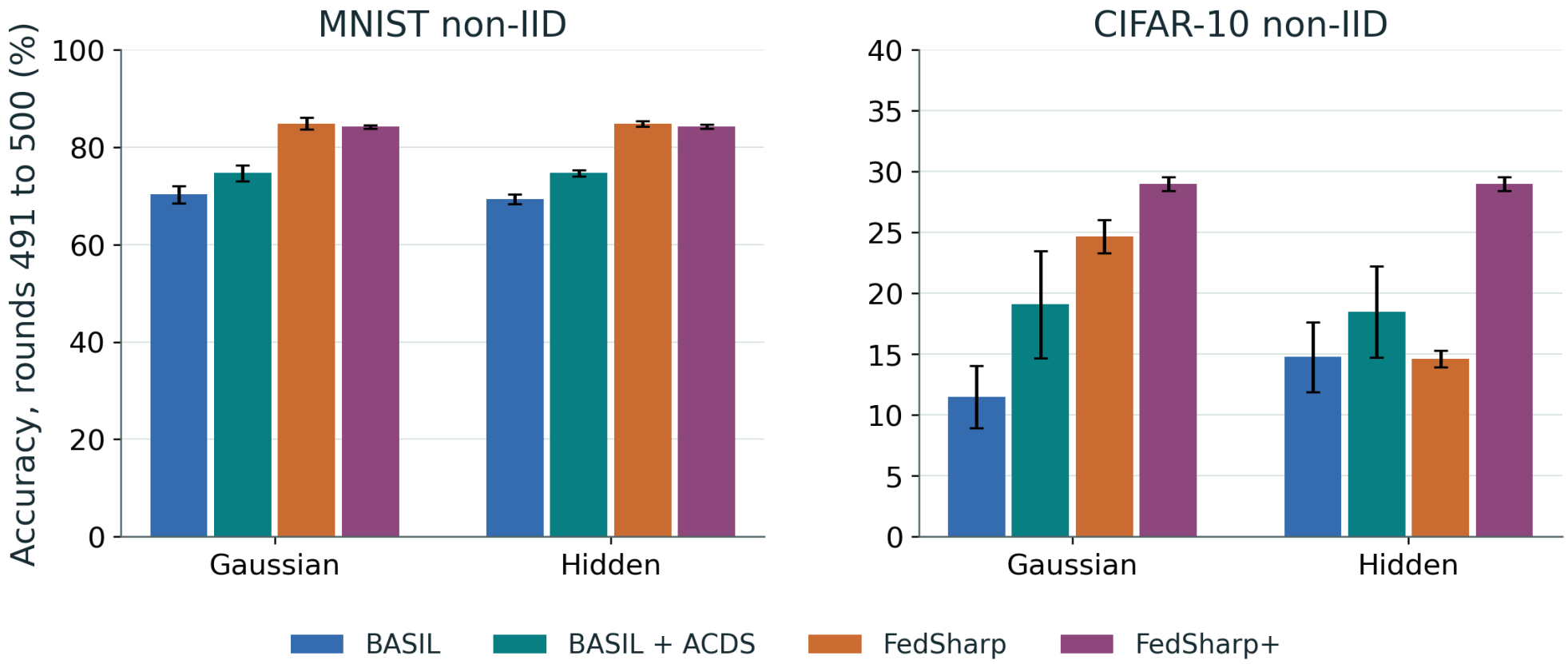
# CIFAR-10 non-IID: accuracy through 500 rounds




**FedSharp+ has the highest joint-impaired mean in this setting.**

Joint = attack + noise. Three seeds; bands = sample SD. Clean curves coincide. Development-holdout accuracy.

# FedSharp+ helps CIFAR-10, but the MNIST gain is absent




**The extension is setting-dependent, rather than a consistent improvement.**

Joint attack + noise only. Endpoint means ± sample SD, three seeds. Note the different dataset scales.

# Mechanism records help explain the remaining tradeoff

**CIFAR-10 non-IID**

99.6%
owned-model admission

**CIFAR-10 non-IID**

28.9%
zero-scale commitment

## Receiver checks often retain a locally established model.

**The checks limit change on local probes; they do not guarantee every class improves.**

# The answer depends on the condition being tested

### FedSharp

Higher means in many joint-impaired cases; BASIL remains stronger in attack-only cases.

### FedSharp+

CIFAR-10 non-IID gains, with no consistent gain over FedSharp on MNIST.

### Clean controls

Highest accuracy in each group, with identical BASIL and FedSharp trajectories.

# Test generalization and isolate each component

| | |
|---|---|
| **Generalization** | Untouched final test data and more seeds. |
| **Attribution** | Separate admission, prior adjustment, repair and commitment. |
| **Deployment** | Adaptive threats, changing topologies and real channel measurements. |

**Keep local probe guarantees separate from claims about population accuracy.**

# Selected references

[1] H. B. McMahan et al. Communication-Efficient Learning of Deep Networks From Decentralized Data. AISTATS, 2017.

[2] A. R. Elkordy, S. Prakash and S. Avestimehr. Basil: A Fast and Byzantine-Resilient Approach for Decentralized Training. IEEE JSAC, 2022.

[3] K. Hsieh et al. The Non-IID Data Quagmire of Decentralized Machine Learning. ICML, 2020.

[4] F. Ang et al. Robust Federated Learning With Noisy Communication. IEEE Transactions on Communications, 2020.

[5] T. Li et al. Federated Optimization in Heterogeneous Networks. MLSys, 2020.

[6] A. K. Menon et al. Long-Tail Learning via Logit Adjustment. ICLR, 2021.

# Thank you

Questions and discussion

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